Electromagnetic Interference And Compatibility: Indian Scenario

Electromagnetic interference (EMI) is a disturbance caused by radiation fields created by electronic devices such as cellular phones or laptops. EMI causes unacceptable degradation of systems or equipment performance. Therefore it's important to develop an effective shielding material to protect the environment and workplace from EMI

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Electromagnetic interference (EMI) is a disturbance caused by radiation fields created by electronic devices such as cellular phones, household devices, communication antennas, etc. The most common example of EMI is the interference of laptop or radio speaker with mobile signals, resulting in the flickering of pictures or buzzing sounds. EMI consists of many undesirable radiated signals, which can cause unacceptable degradation of systems or equipment performance.

These problems, if left unattended, can cause severe damage to communication systems and electronic devices. Therefore it's important to develop an effective shielding material to protect the environment and workplace from EMI due to unwanted electromagnetic waves. EMI shielding refers to the reflection and adsorption of electromagnetic radiation by a material.

Causes of EMI

Electromagnetic radiation emitted by electronic devices is the cause of EMI. The most common cause of EMI is electrostatic discharge (ESD).

The sources of EMI can be generally separated into the following categories:

- Incidental interference
- Environmental interference
- Band congestion
- Inter-modulation distortion
- External noise
- Adjacent channel interference
- Intentional interference (jamming)

EMI may transfer from source to victim by induction, conduction, and radiation.

The intentional release of energy to cause